

Blockchain-Enabled Beneficiary Registry Management and Fraud Detection for Public Distribution System (PDS)

Problem Statement

- The Public Distribution System (PDS) is a critical welfare mechanism that provides subsidized food grains and essential commodities to eligible beneficiaries under the National Food Security Act (NFSA).
- The effectiveness of PDS depends upon maintaining an accurate transparent and continuously updated beneficiary database.
- The presence of ghost beneficiaries including duplicate fake deceased migrated and otherwise ineligible beneficiaries remains a significant challenge affecting the efficiency and integrity of the PDS ecosystem.
- Beneficiary records may become inaccurate due to delayed updates manual data entry errors inadequate verification mechanisms duplication across databases migration of beneficiaries and non-reporting of deaths or changes in eligibility status.
- In Jammu & Kashmir the PDS serves a large and geographically dispersed population across urban rural hilly remote and border areas making beneficiary verification and database maintenance a complex administrative task.
- Frequent demographic changes such as births deaths migration family bifurcations and changes in economic status require continuous updating and validation of beneficiary records.
- The existence of ghost beneficiaries leads to diversion and misuse of subsidized food grains resulting in leakage of public resources and financial losses to the government.
- Duplicate and fraudulent beneficiary records distort food grain allocation demand forecasting procurement planning and subsidy expenditure calculations.

- Existing centralized databases do not always provide a complete immutable and auditable history of beneficiary creation modification verification and deletion activities.
- Multiple agencies and systems involved in beneficiary lifecycle management often operate independently creating challenges in data synchronization validation and cross-verification.
- Although Aadhaar seeding e-KYC authentication and ePoS-based distribution have improved transparency challenges remain in identifying duplicate records detecting fraudulent activities and maintaining a trusted beneficiary registry.
- The absence of a tamper-proof audit trail makes it difficult to identify unauthorized modifications establish accountability and investigate fraud-related incidents.
- Lack of a single trusted source of beneficiary information can result in data inconsistencies operational inefficiencies and delays in detecting ineligible beneficiaries.
- There is a need for a secure transparent and immutable mechanism that can maintain a trusted record of all beneficiary-related transactions and lifecycle events.
- A blockchain-enabled beneficiary registry can provide a decentralized and tamper-proof platform for beneficiary verification fraud detection and lifecycle management thereby reducing ghost beneficiaries improving database accuracy minimizing subsidy leakage and strengthening trust in the Public Distribution System.